

FREE ELECTIVE 4 (FRE413)

PRELIMINARY PARALLEL ACTIVITY

Design Your Data Analytics Application

Course: FREE ELECTIVE 4 – Data Analytics with Flutter

Coverage: Weeks 2–4

Total: 100 Points

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Overview

For this Preliminary Parallel Activity, you will analyze how data can be used in an application.

Use your **existing project** as the basis of this activity. You will identify the data used by your project, classify its data types and sources, examine possible data-quality problems, design a simple analytics interface, and explain how a database and API could support the application.

This is an **individual activity**. If your existing project was developed by a group, each member must prepare and submit their own analysis.

General Instructions

1. Use your **existing project** as the basis of your answers.
2. Complete all seven parts of the activity.
3. Keep your answers clear and concise.
4. The UI sketch may be created:
 - By hand on paper
 - Using Microsoft Word or PowerPoint
 - Using any available design application
5. Programming or Flutter coding is **not required** for this activity.
6. Internet access is not required for most parts of the activity.

7. Your work may initially be handwritten or prepared digitally.
 8. Before submission, combine your work into **one PDF file**.
 9. Include your **Name, Year/Section, Project Title, and Date**.
 10. Submit the PDF according to the schedule provided by your instructor.
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PART 1 – PROJECT AND DATA IDENTIFICATION

15 Points

Identify your existing project.

Provide the following:

Project Title:

Purpose of the Project:

Briefly explain what problem your project solves.

Then identify at least **five (5) pieces of data** used by your project.

Complete the table.

No.	Data	Description
1		
2		
3		
4		
5		

Example

No.	Data	Description
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1	Product Name	Name of the product stored in the system
2	Quantity	Number of available products

Scoring

- Project title and purpose – **5 points**
- Five appropriate pieces of data – **10 points**

PART 1 TOTAL: 15 POINTS

PART 2 – DATA TYPE CLASSIFICATION

15 Points

Classify the five pieces of data identified in Part 1.

Use the following classifications:

- **Structured Data**
- **Semi-Structured Data**
- **Unstructured Data**

Complete the table.

Data	Data Type	Explanation

Your explanation should briefly state **why** the data belongs to that classification.

Example

Data	Data Type	Explanation
Product Name	Structured	It can be stored in a predefined database field.
Product Image	Unstructured	An image does not follow a row-and-column data format.

Scoring

Each correctly classified and explained item is worth **3 points**.

5 items × 3 points = 15 points

PART 3 – DATA SOURCES

15 Points

Identify where the data in your application comes from.

For each of your five pieces of data, identify whether its source is:

Internal or External

and, where applicable:

Primary or Secondary

Complete the table.

Data	Internal / External	Primary / Secondary	Source

Example

Data	Internal / External	Primary / Secondary	Source
Customer Name	Internal	Primary	Registration Form
Weather	External	Secondary	Weather API

Scoring

Each properly identified data source is worth **3 points**.

5 items × 3 points = 15 points

PART 4 – DATA QUALITY

15 Points

Identify **three (3) possible data-quality problems** that could occur in your project.

For each problem:

1. Identify the problem.
2. Give an example.
3. Explain how the application could prevent or correct it.

Complete the table.

Data Quality Problem	Example	Prevention / Solution

Example

Data Quality Problem	Example	Prevention / Solution
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Invalid quantity	Quantity = -20	Do not allow quantities below 0.
Missing information	Product name is empty	Make the product name required.

Consider characteristics of good data such as:

- Accurate
- Complete
- Consistent
- Relevant
- Timely

Scoring

Each problem is worth **5 points**:

- Appropriate problem – 1 point
- Clear example – 2 points
- Appropriate prevention/solution – 2 points

3 problems × 5 points = 15 points

PART 5 – ANALYTICS DASHBOARD UI SKETCH

15 Points

Create a simple sketch showing how data from your existing project could be presented in an **Analytics Dashboard**.

Your sketch must contain at least:

- Application/Project Name
- At least **three (3) summary cards or information areas**
- At least **one (1) area for displaying records or information**

Example

+-----+

INVENTORY ANALYTICS

+-----+ +-----+ +-----+		
Total	Low Stock	Categories
Products	Items	
120	8	5
+-----+ +-----+ +-----+		

PRODUCT RECORDS

Product	Quantity	Price
Laptop	10	35000
Mouse	25	500
Keyboard	15	1200

+-----+

Your design does not need to look exactly like the example.

The dashboard should reflect the type of data used by **your own project**.

Scoring

Requirement	Points
Project/Application title	2
Three appropriate summary areas	6
Data/records display area	4
Organization and readability	3
TOTAL	15

PART 6 – DATABASE PLAN

15 Points

Explain how a database could be used in your existing project.

Create at least **one database collection or table**.

Provide:

Collection/Table Name:

Identify at least **five (5) fields**.

Field	Purpose	Example Value

Example

Collection/Table: products

Field	Purpose	Example Value
productName	Stores product name	Laptop
quantity	Stores available quantity	10
price	Stores product price	35000

Then briefly answer:

Why does your application need a database?

Scoring

- Appropriate collection/table – **2 points**
- Five appropriate fields – **10 points**
- Explanation of database purpose – **3 points**

PART 6 TOTAL: 15 POINTS

PART 7 – API APPLICATION

10 Points

Identify **one type of external information** that could improve your existing project.

Examples include:

- Weather information
- Maps and location
- Currency exchange rates
- News
- Product information
- Public statistics
- Other appropriate external information

Complete the following:

External Information Needed:

Possible API / Data Source:

How will this information help your application?

How could the information be displayed in your application?

Example

Project: Delivery Management System

External Information Needed: Weather information

Possible API / Data Source: Weather API

Purpose: Weather information can help determine whether weather conditions may affect delivery operations.

Display: The current weather condition can be displayed on the delivery dashboard.

Scoring

- Appropriate external information – **2 points**
- Appropriate API/data source – **2 points**
- Clear explanation of its purpose – **3 points**
- Appropriate application/display of the information – **3 points**

PART 7 TOTAL: 10 POINTS

SCORING SUMMARY

Part	Component	Points
1	Project and Data Identification	15
2	Data Type Classification	15
3	Data Sources	15
4	Data Quality	15
5	Analytics Dashboard UI Sketch	15
6	Database Plan	15
7	API Application	10
TOTAL		100 POINTS

FINAL SUBMISSION CHECKLIST

Before submitting, make sure your work contains:

- Name
 - Year and Section
 - Project Title
 - Part 1 – Project and Data Identification
 - Part 2 – Data Type Classification
 - Part 3 – Data Sources
 - Part 4 – Data Quality
 - Part 5 – Analytics Dashboard UI Sketch
 - Part 6 – Database Plan
 - Part 7 – API Application
 - All pages combined into one PDF file
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IMPORTANT REMINDER

This activity must be based on **your own existing project and its data**.

Your answers, database plan, dashboard sketch, and API application should match the purpose and data requirements of your selected project.

The goal is to demonstrate how the concepts covered during the Preliminary Period can be applied to an actual application.

TOTAL SCORE: 100 POINTS